

Smart White Cane with AI Navigation and Haptic Feedback

An IOT Based Assistive Navigation System for Visually Impaired Users



PH 3042 – Robotics and Automation

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Abstract

Visual impairment significantly affects mobility, independence and the ability to navigate safely in both familiar and unfamiliar environments. Traditional white canes, although widely used as primary mobility aids, mainly detect ground level obstacles through physical contact and often fail to identify overhead hazards or dynamic obstacles which can compromise user safety (Dakopoulos & Bourbakis, 2010). Recent advancements in sensors, embedded systems and artificial intelligence have enabled the development of smart assistive technologies capable of providing real time navigation guidance and improving situational awareness (Katzschmann et al., 2018). This project presents the design and implementation of a **Smart White Cane** that integrates ultrasonic obstacle detection, directional haptic feedback, AI based indoor navigation and mobile assisted outdoor navigation to support visually impaired users.

The system architecture is built around an **ESP32S3 microcontroller** which manages sensor data processing, haptic feedback and wireless communication with a mobile application. Obstacle detection is achieved using an ultrasonic sensor mounted on the cane to measure the distance to obstacles ahead of the user and this technique widely used in assistive navigation systems due to its reliability and low cost (Bousbia-Salah et al., 2011). Three vibration motors are positioned around the cane handle at left, bottom and right positions. The motors provide directional and intensity based haptic feedback which allows users to perceive both the location and proximity of obstacles without relying on auditory cues and preserves environmental awareness.

Indoor navigation is facilitated using an **ESP32-CAM module** integrated into the lower section of the cane. The camera continuously captures images of the floor area in front of the user and detects predefined visual markers placed along indoor pathways using a lightweight embedded machine learning model. Marker based navigation systems have been widely explored for indoor guidance where GPS signals are unavailable (Kulyukin et al., 2008). Each detected marker corresponds to specific navigation instructions such as forward, left turn or right turn. Feedback is provided through haptic vibration patterns or audio cues which enables the user to safely follow complex indoor routes such as corridors or hallways even in the absence of GPS signals.

Outdoor navigation is supported through a mobile application connected via Wi-Fi. The application utilizes GPS based mapping services to determine the location of users and calculate optimal routes to selected destinations. Voice guided

instructions, including turn directions and distance updates are transmitted to the cane through an integrated speaker which provides continuous navigation assistance in outdoor environments (Ahmetovic et al., 2016). The system is powered by two 18650 lithium-ion batteries connected in parallel and regulated using a boost converter to deliver a stable 5V output to the sensors, microcontrollers and communication modules.

Extensive system analysis and testing demonstrate that the Smart White Cane can reliably detect obstacles ranging from low lying hazards to overhead obstructions and provides real time haptic alerts that enhance safety and situational awareness. The AI based indoor navigation module successfully guides users along predefined paths while the mobile assisted outdoor navigation system facilitates real time route guidance across diverse environments. By integrating sensing, embedded processing, machine learning and mobile connectivity, the proposed system significantly improves the mobility, safety and confidence of visually impaired users.

The innovation of this Smart White Cane lies in its hybrid navigation approach which combining tactile obstacle alerts with AI assisted guidance and ensures a seamless transition between indoor and outdoor navigation. Unlike conventional canes, the proposed system proactively detects obstacles, reduces collision risk and provides intuitive and non-intrusive guidance. Future enhancements could include advanced object recognition cloud-based AI processing and improved indoor localization to further augment usability and reliability.

In conclusion, the Smart White Cane represents a practical and scalable solution that leverages modern robotics and AI technologies to empower visually impaired individuals. By improving situational awareness, route guidance and independence, this device has the potential to enhance the quality of life and safety of users while demonstrating the transformative impact of assistive technology in daily life.

References

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